INTERNSHIP REPORT

On

Full Stack Web Development with AI Tools

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Year: 2025–2026

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Year 2025–26

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This is to certify that Internship report entitled

"Full Stack Web Development with AI Tools"

is submitted as partial fulfilment of

curriculum of the T.E. of information technology

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Chapter 1

Internship Place Details

Name of Company: Edunet Foundation
Place: Online / Remote
Area of Internship: Web Development
Duration: 4 Weeks

Company Background

Edunet Foundation is a well-known organization dedicated to enhancing the technical and professional skills of students across India. The foundation focuses on bridging the gap between academic learning and industry requirements by providing hands-on training in emerging technologies such as web development, artificial intelligence, cloud computing, and data science.

The organization collaborates with industry experts and educational institutions to design structured training programs that are aligned with current market demands. Through its initiatives, Edunet Foundation aims to improve students' employability by equipping them with practical knowledge, problem-solving abilities, and industry-relevant experience.

Activities Performed

- During the 4-week internship, various learning and development activities were conducted to provide both theoretical knowledge and practical exposure. These activities are described below:
- Technical Training Sessions The internship began with a series of

technical training sessions focused on web development fundamentals. These sessions covered essential topics such as:

HTML, CSS, and JavaScript basics

Responsive web design

Introduction to frameworks and tools

1.1 Summary

Overall, the internship at Edunet Foundation was a valuable learning experience that enhanced both technical and professional skills. It provided a balanced combination of training, practical implementation, and career preparation, making it highly beneficial for future career growth.

Chapter 2

Introduction

2.1 Problem Statement

The rapid growth of online blogs makes it difficult for users to find relevant and personalized content. Traditional blogging platforms lack effective recommendation systems, leading to poor user engagement. This project aims to develop a blog website integrated with a KNN-based recommendation system to provide personalized blog suggestions.

2.2 Objectives

- Design and develop a dynamic blogging website
- Enable users to create, read, update, and delete blog posts
- Implement a KNN-based recommendation system for personalized blog suggestions
- Improve user engagement by suggesting relevant content
- Understand the integration of machine learning with web applications
- Gain practical experience in full-stack web development

2.3 Motivation / Scope

With the increasing amount of online content, users often struggle to find blogs that match their interests. Traditional blogging platforms do not provide personalized recommendations, which reduces user engagement.

This project is motivated by the need to enhance user experience by integrating machine learning techniques to deliver relevant and personalized blog suggestions.

The project focuses on developing a blogging website with basic functionalities like creating, editing, and viewing blogs. It also includes a KNN-based recommendation system to suggest relevant content to users. The system can be further enhanced by adding features like comments, likes, user profiles, and advanced recommendation algorithms for better accuracy and scalability.

Chapter 3

Project Completed

3.1 Number of Modules

- Login and Registration Module
- Admin Panel
- Home Page
- About us
- Contact us
- Add blog,Edit Blog
- Blogs
- Django administration

3.2 Login Module

3.2.1 Features

- User-friendly interface
- Login Sign Up system
- AI-based attrition prediction
- Uses machine learning

3.2.2 Description

This screenshot represents the loginpageThe Login Page of Blogsphere – A Web-Based Blog Platform serves as the entry point for registered users to securely access their accounts and personalized blog experience. It is designed with a simple and user-friendly interface to ensure smooth authentication.

The page consists of the following components:

Navigation Bar: Located at the top, it includes links such as Home, Blog, About, and Contact, along with options for Login and Register. This allows users to easily navigate across different sections of the platform. Login Form: Positioned centrally, the login form provides a focused area for user authentication. It includes: Username Field: Allows users to enter their registered username. Password Field: A secure input field where the password is masked for privacy. Login Button: A prominent button that submits the entered credentials for authentication. Upon successful login, users are redirected to their personalized dashboard or homepage. Remember Me Option: A checkbox that allows users to stay logged in on the device, enhancing convenience for frequent users. Create Account Option: A link for new users to register on the platform, ensuring easy onboarding. Forgot Password Feature: Provides a recovery option for users who have forgotten their passwords, improving accessibility and user support. Footer Section: Displays basic information about the platform, such as copyright details and social media links, contributing to the overall professional design. Overall Functionality

This page plays a critical role in enabling secure access to the platform. By authenticating users, it allows them to interact with personalized features such as content recommendations, blog creation, and curated feeds—directly supporting the platform's goal of solving content discovery challenges.

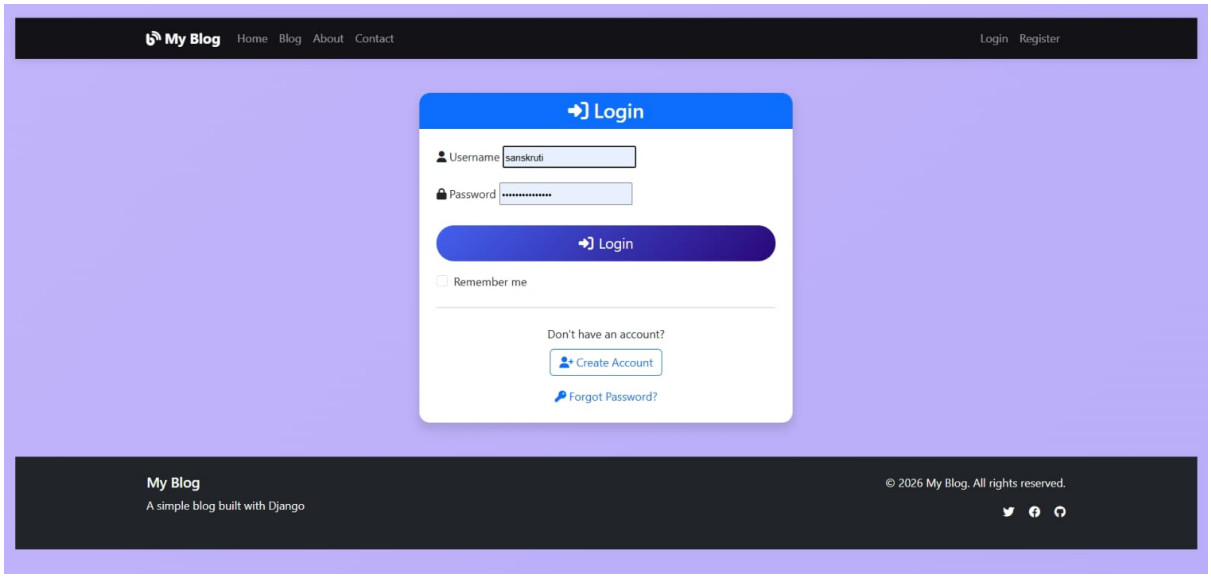

Figure 3.1: Home Page

3.3 Create and Edit Profile

3.3.1 Features

- User friendly interaction
- Edit the profile
- Create the profile

3.3.2 Description

The User Registration and Profile Management module is an important part of the blog website that enables users to create accounts and manage their personal information efficiently. This module consists of two main functionalities: account creation (registration) and profile editing. It ensures that users can securely register on the platform and personalize their profiles according to their preferences.

The Create Account page provides a structured form where new users can enter their details such as username, email address, first name, last name, and password. The system ensures proper validation of user inputs, including password strength requirements such as minimum length, avoidance of common passwords, and confirmation matching. This helps in maintaining security and protecting user data. The interface is designed to be simple and intuitive, guiding users through the registration process with clear instructions.

The Edit Profile page allows registered users to update their personal information such as username, email ID, name, phone number, and bio. This feature enhances user flexibility by allowing them to modify their details whenever required. The inclusion of a bio field enables users to describe themselves, making the platform more interactive and personalized. The page also includes action buttons like "Save Changes" and "Cancel," giving users full control over their updates.

Both pages are designed with a clean and responsive layout to ensure compatibility across different devices. Proper form alignment, spacing, and visual elements improve usability and overall user experience. This module plays a key role in maintaining user identity, enhancing personalization, and supporting secure access to the blogging platform

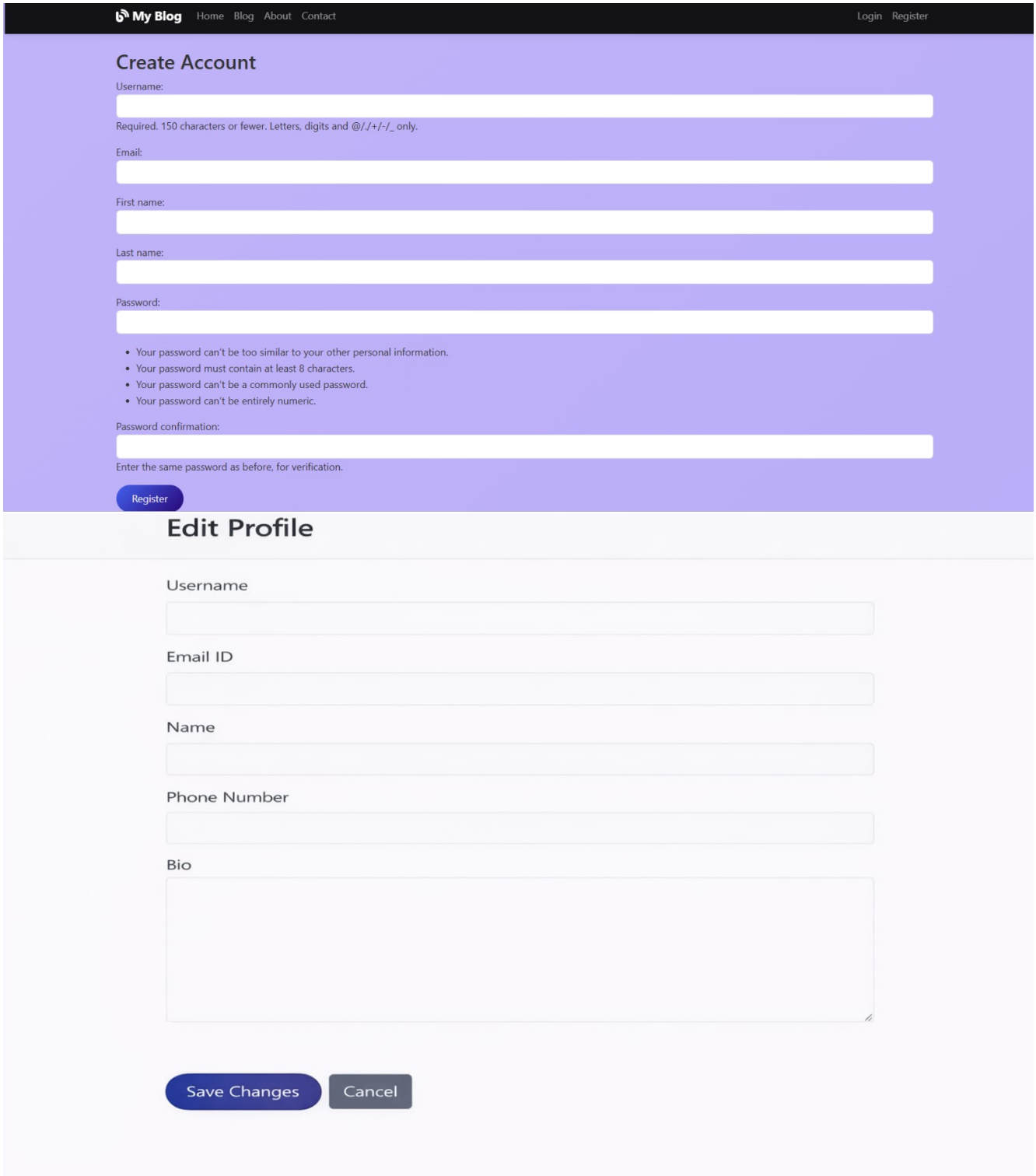

Figure 3.2: Create and edit profile Page

3.4 Admin Panel

3.4.1 Description

The Django Administration Pages provide a built-in interface that allows administrators to manage the entire blog website efficiently without directly interacting with the database or writing additional code. This admin panel is one of the most powerful features of the Django framework, offering a secure and user-friendly environment for handling backend operations.

In this project, the Django admin interface is used to manage important data such as user accounts, blog posts, categories, and other related information. The admin panel is accessible only to authorized users (superusers or staff members), ensuring that sensitive data remains protected. Through this interface, the administrator can perform operations like adding new blog posts, editing existing content, deleting unwanted data, and monitoring user activities.

The admin dashboard displays all registered models in a structured format, making it easy to navigate between different sections. Each model provides options to view records in a list format, apply filters, search for specific entries, and perform bulk actions. The interface is automatically generated by Django but can also be customized according to project requirements.

One of the key advantages of the Django admin panel is its ability to simplify content management. Instead of creating separate interfaces for backend operations, the admin panel provides ready-to-use functionality, saving development time and effort. It also ensures consistency and reliability in data handling.

Overall, the Django Administration Pages play a crucial role in maintaining the blog website by providing efficient data management, secure access control, and easy monitoring of system activities. It enhances productivity and ensures smooth functioning of the application

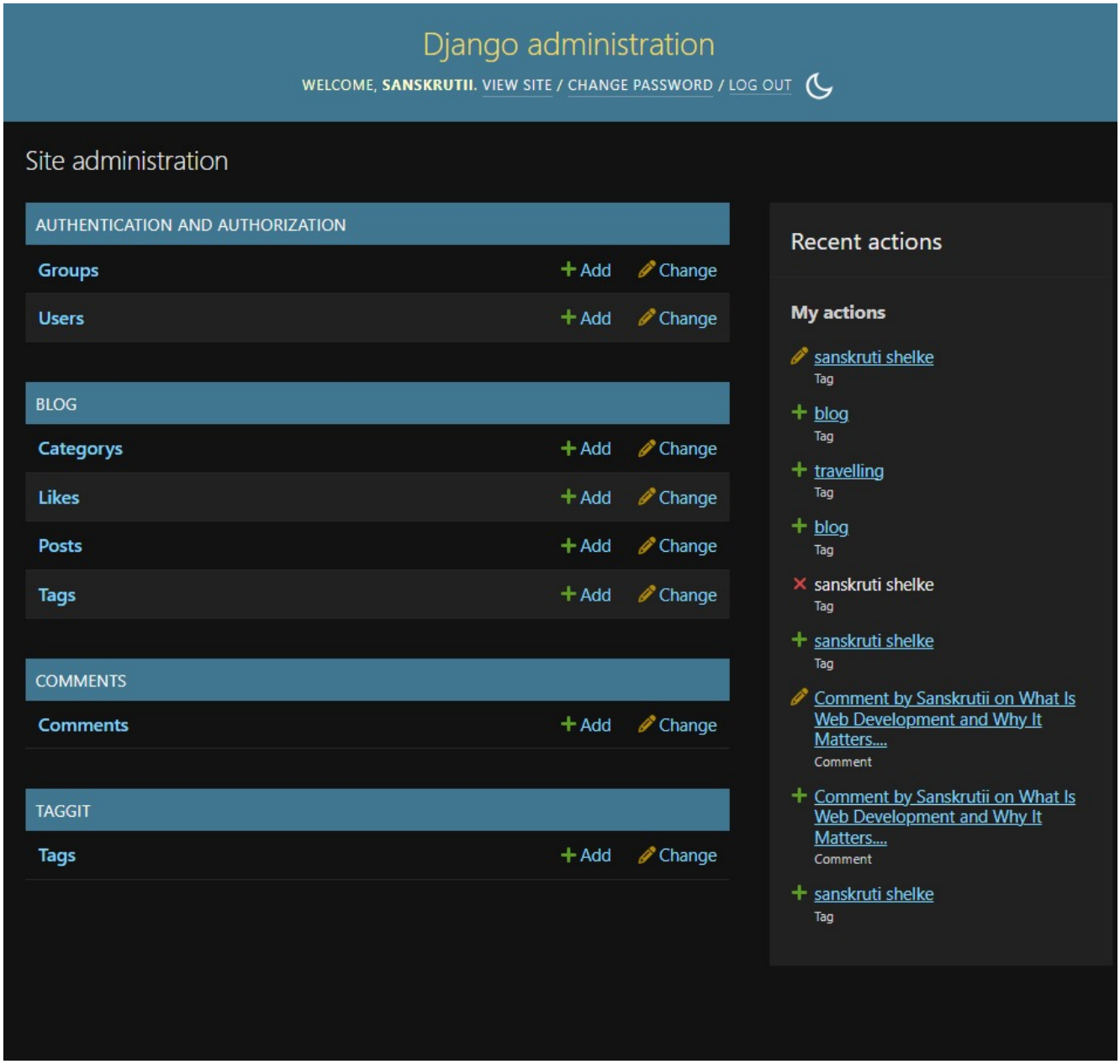

Figure 3.3: Admin Panel

3.5 Home Page

3.5.1 Features

- Aggregation in One Place
- Personalized Discovery
- Categorization Quality Curation
- User Engagement Contribution
- Efficient Navigation

3.5.2 Description

This home page is the user's first interaction with your aggregated, personalized blogosphere. It replaces scattered, non-personalized browsing with a one-stop, tailored content discovery engine.

1. Aggregation in One Place

Instead of visiting multiple websites, users land here and see all blog content (recent posts, recommended posts) in a single, clean interface. The navigation (Home, Blog, About, Contact) keeps everything organized.

2. Personalized Discovery

The "Recommended For You" section explicitly tackles "lack of personalization" and "inefficient discovery".

"Posts tailored to your interests based on your reading history"

This means your backend tracks user behavior and suggests relevant content automatically—no manual searching across scattered sources.

3. Categorization Quality Curation

· Recent Posts with titles like "Python..." and "What Is Web Development..." show categorized, curated content. · Features like Categories Tags, Comments System, and CRUD operations ensure posts are structured and high-quality, not random or low-effort.

4. User Engagement Contribution

· User Registration Authentication + Write a Post allow users to become creators, enriching the platform's content pool. · The "Welcome to My Blog" intro and About This Blog section build trust and context, helping users understand they're in a dedicated content hub.

5. Efficient Navigation

· Recent Posts give instant access to fresh content. · A logged-in user sees a personalized experience, reducing the time spent hunting for relevant reads.

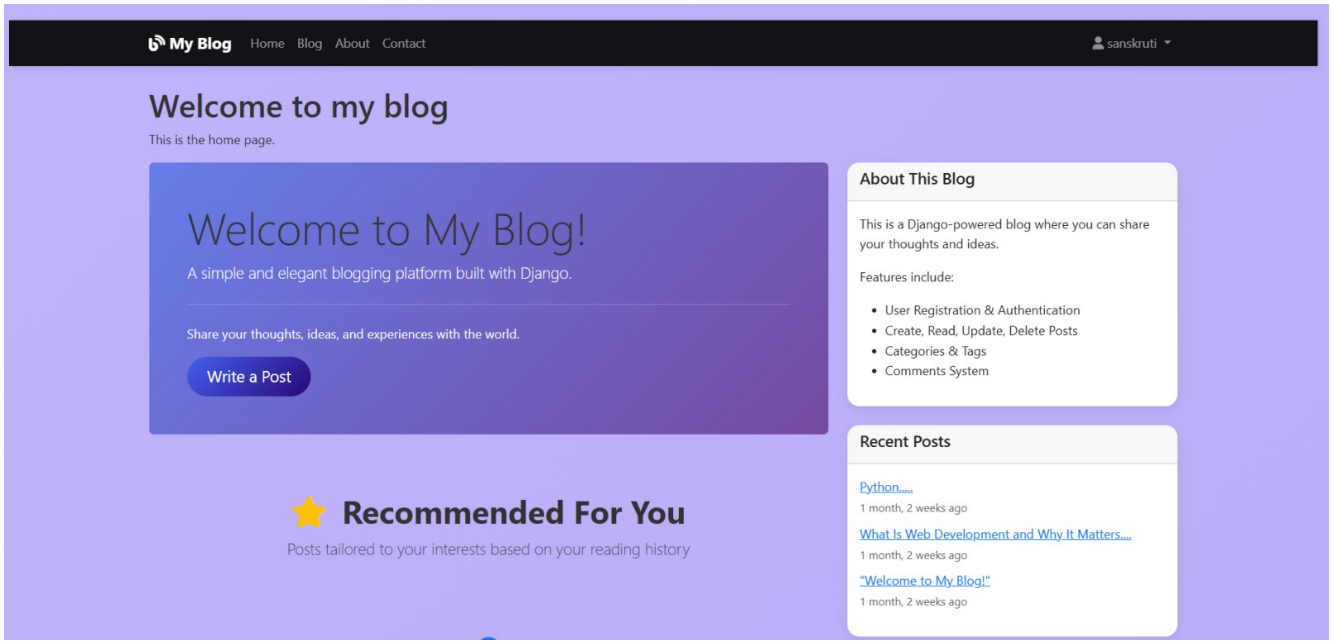

Figure 3.4: Home Page

3.6 About My blogs

3.6.1 Features

- Establishing Credibility Quality Assurance
- Encouraging User Participation to Enrich Content
- Building a Community Around Shared Interests
- Quick Stats for Transparency Growth Tracking
- Reinforcing the "One-Stop Hub" Concept

3.6.2 Description

This page plays a crucial supporting role in building trust, transparency, and community, which are essential for a successful content aggregator.

Here's how this page contributes to solving your problem:

1. Establishing Credibility Quality Assurance

Your problem states that content is scattered and often lacks quality. This page communicates the platform's mission ("to create a platform where everyone can express themselves freely") and its Django-powered foundation. By being transparent about purpose and technology, users are more likely to trust that the aggregated content here is curated and meaningful—not random or low-effort.

2. Encouraging User Participation to Enrich Content

A content aggregator needs a steady stream of high-quality posts. The "Want to contribute? Join our community of writers!" and "Write a Post" call-to-action directly solve the "lack of relevant content" issue by inviting users to become creators. More contributors = richer, more diverse content for everyone.

3. Building a Community Around Shared Interests

The problem mentions inefficient discovery because content is scattered across multiple websites. By highlighting "Join thousands of writers and readers sharing knowledge every day" and "Our Community", Blogosphere positions itself as the single destination where both readers and writers gather—eliminating the need to browse multiple external blogs.

4. Quick Stats for Transparency Growth Tracking

The Quick Stats section shows:

· Total Posts: 0 · Total Users: 0 · Total Comments: 0 · Categories: 0

This may look empty now, but it sets user expectations that the platform is growing. As these numbers increase, they become social proof—demonstrating that Blogosphere is an active, content-rich hub worth returning to.

5. Reinforcing the "One-Stop Hub" Concept

The footer ("A simple blog built with Django") and consistent branding (My Blog) remind users that everything lives under one roof. No need to jump between external sites—Blogosphere aggregates, categorizes, and recommends everything from here.

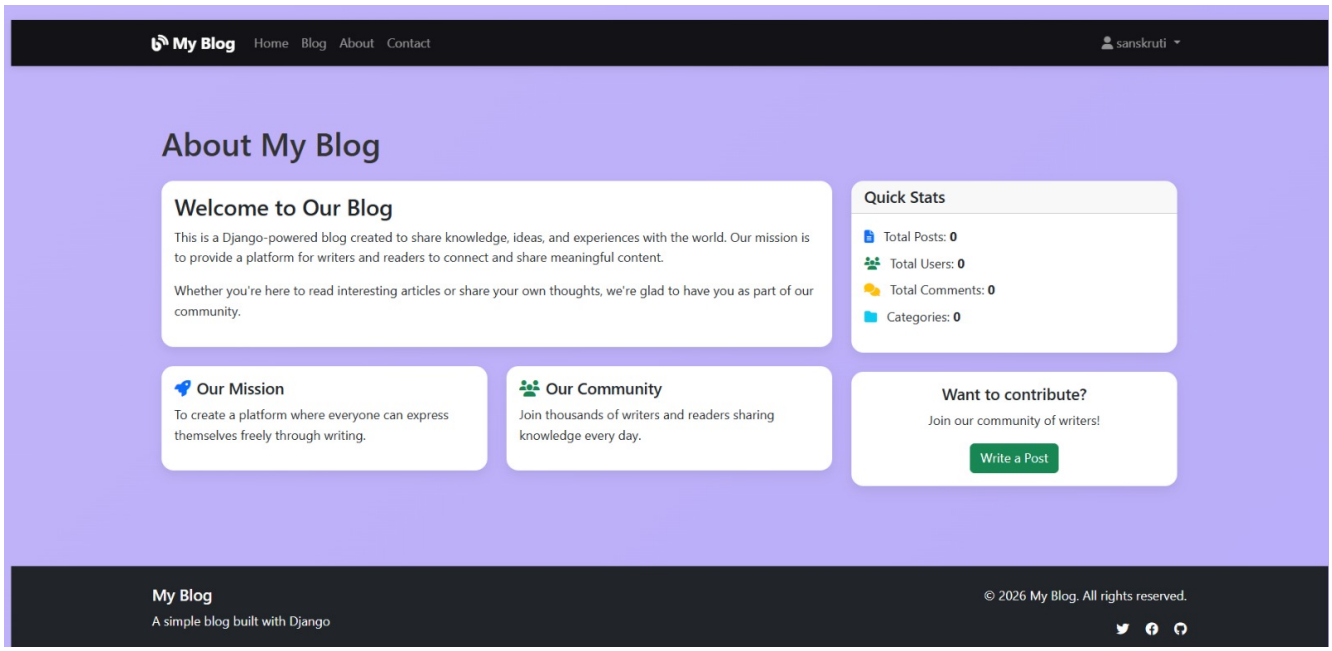

Figure 3.5: About Blog Page

3.7 Contact Us

3.7.1 Features

- Effective communication between users an admin
- Well organized layout
- Clean and visually appealing
- Enhance usability

3.7.2 Description

The Contact Us page is an essential component of the blog website that facilitates effective communication between users and the website administrator. It is designed to provide a simple, structured, and user-friendly interface through which users can submit their queries, feedback, suggestions, or report issues. This page plays a crucial role in improving user engagement and maintaining transparency by allowing users to directly connect with the system.

The page consists of a well-organized layout divided into two main sections. The first section includes a contact form where users can input their details such as name, email address, subject, and message. This form is designed with proper input fields and placeholders to guide users in entering accurate information. The second section displays important contact information, including the official email address, response time, and support working hours, which helps users understand when and how they can expect a reply.

The design of the page is clean and visually appealing, ensuring a smooth user experience. It uses a responsive layout, making it accessible on different devices such as desktops, tablets, and mobile phones. The page also includes an additional guidance section that encourages users to check existing resources like blog posts or FAQs before submitting their queries, which helps reduce unnecessary messages and improves efficiency.

Overall, the Contact Us page enhances the usability and professionalism of the blog website by providing a reliable communication channel. It ensures that user concerns are addressed effectively and contributes to building trust and interaction between users and the platform.

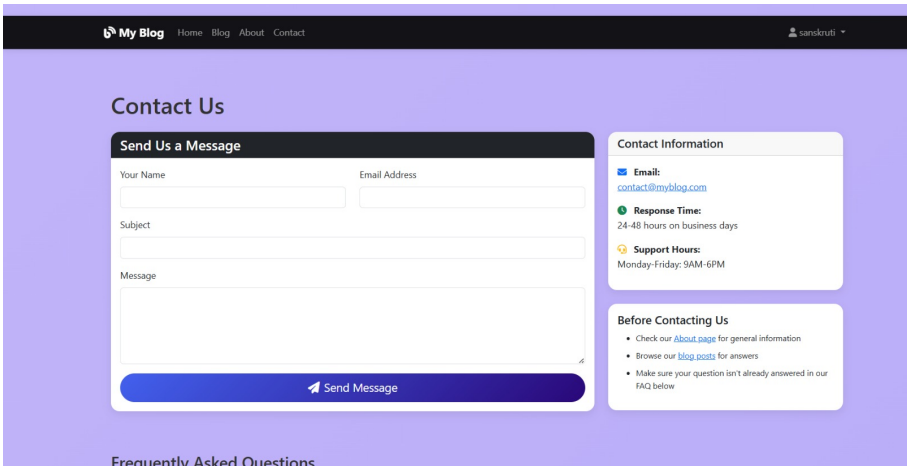

Figure 3.6: Contact Us Page

3.8 Create Blogs

3.8.1 Features

- Create new blogs
- Create content
- Logout functionality

3.8.2 Description

The Create New Post page in Blogsphere – A Web-Based Blog Platform provides users with an intuitive interface to compose, manage, and publish blog content efficiently. This page is designed to streamline the content creation process while ensuring proper organization and categorization of posts.

It consists of the following key components:

Title Field: A mandatory input field where users enter the title of their blog post. This helps in identifying and displaying the post across the platform. Content Field: A large text area where users can write the main body of the blog. It supports detailed content creation, allowing users to share information, ideas, or insights. Category Selection: A dropdown menu that allows users to assign the post to a specific category or domain. This helps in organizing content and improves discoverability within the platform. Tags Section: Users can select existing tags or add new ones. Tags help in refining search results and enhancing content recommendation by associating posts with relevant keywords. Image Upload: This feature allows users to upload a cover image for the blog post, improving visual appeal and engagement. Status Option: Users can choose the status of the post (e.g., Draft or Published). This enables them to save unfinished work and publish it later. New Tags Input: A field for adding custom tags, entered as comma-separated values, enabling flexibility in content classification. Action Buttons: Publish Post: Submits the blog for publishing or saving based on the selected status. Cancel: Discards the current changes and exits the page. Overall Functionality

This page plays a crucial role in the platform by enabling content creators to contribute blogs that are automatically categorized and integrated into the system. Combined with the aggregator and recommendation features, it ensures that high-quality, relevant content is structured and easily accessible to users.

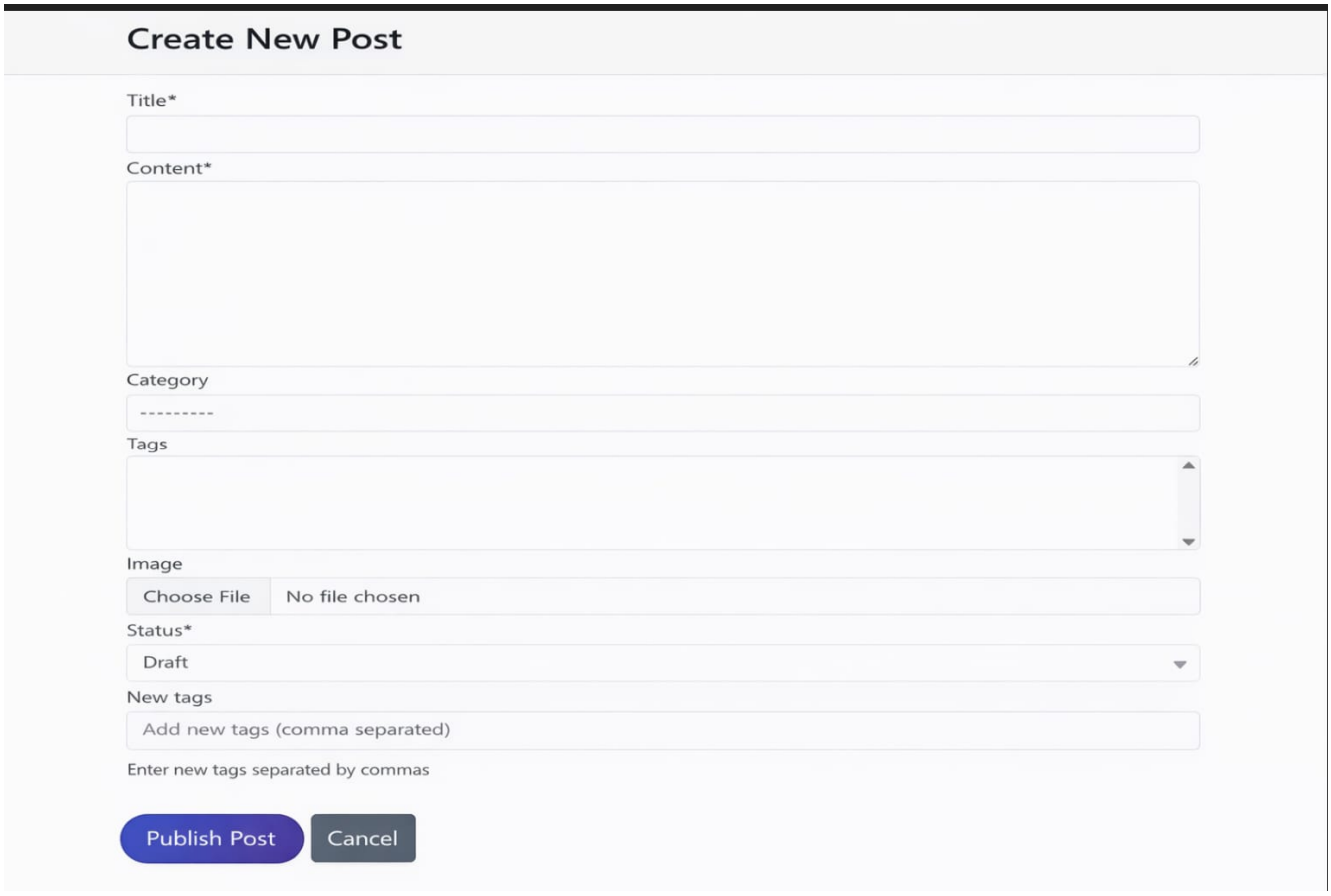

Figure 3.7: Create Blog

3.9 Edit Blog

3.9.1 Features

- Error handling
- Structured content creation
- Logout functionality
- Centralized Publishing

3.9.2 Description

This page empowers bloggers to contribute clean, categorized content — which your backend then collects, filters, and recommends — directly solving the "inefficient discovery" problem stated in your project.

1. Structured Content Creation · Fields like Title, Content, Category, and Tags ensure every post is properly labeled. This allows your platform to

automatically classify posts (e.g., “Python Tips”) and later recommend them to users interested in similar topics.

2. Error Handling Quality Enforcement · The example code snippet with try-except reflects a best-practice tip, but in the platform’s logic, validation would prevent incomplete or malformed posts from entering the aggregation pool — maintaining content quality.

3. Centralized Publishing · Instead of bloggers posting across multiple external websites, Blogosphere becomes the single hub. Posts created/edited here feed directly into your domain-based aggregator.

4. Personalization Readiness · Categories and tags become the metadata that powers personalized recommendations, helping users discover relevant content without manual searching across scattered sources.

5. Media Status Management · The image upload and status toggle (Published/Draft) give creators control, while ensuring only finished, high-quality posts are surfaced to end-users.

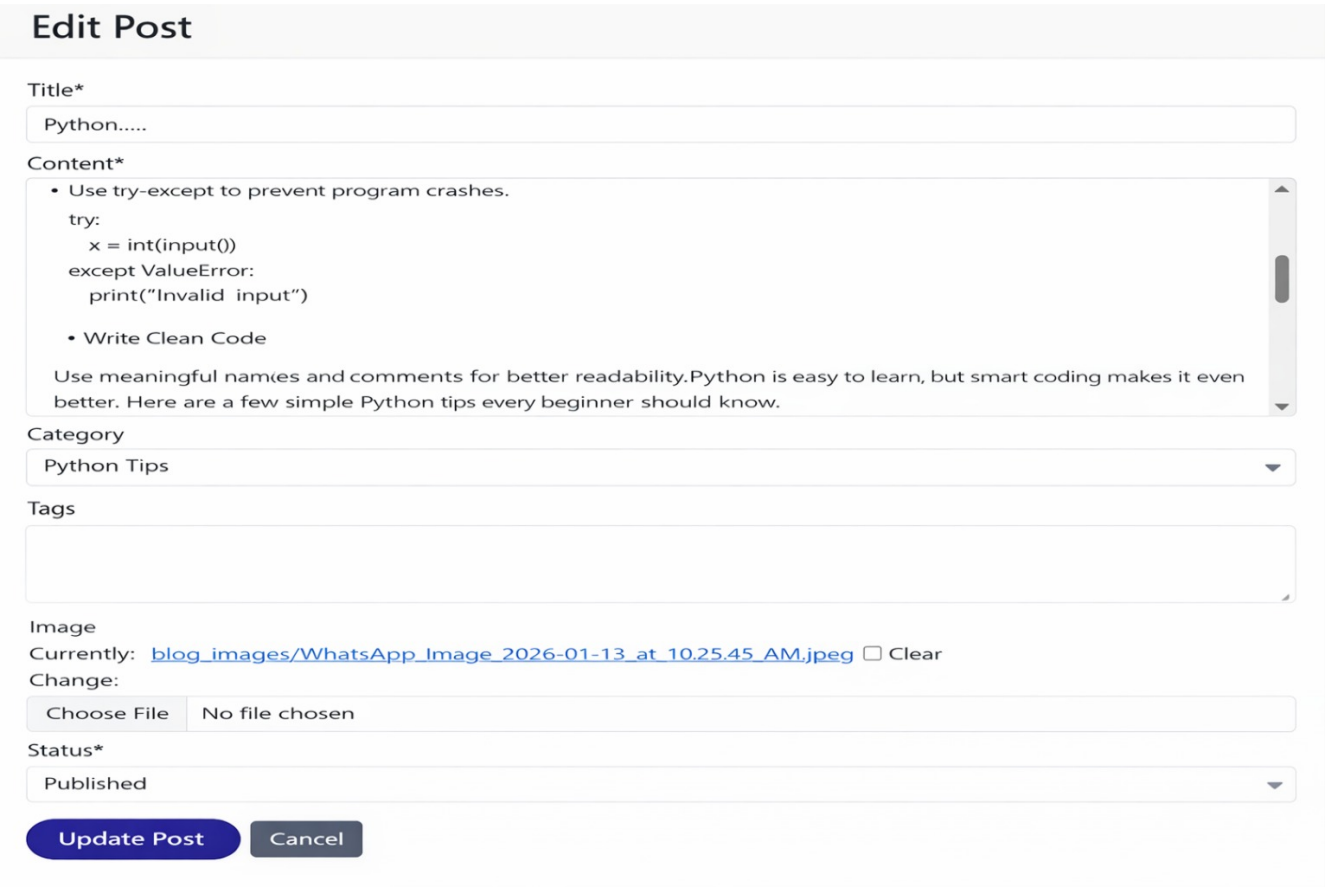

Figure 3.8: Edit Blog Page

3.10 Blog Post

3.10.1 Features

- Featured image and title section
- Blog Content Area
- Interaction Buttons
- Comments Section
- Related posts section

3.10.2 Description

The Blog Post Detail Page in Blogsphere – A Web-Based Blog Platform is designed to display the complete content of a selected blog post in an interactive and user-friendly manner. It allows users to read, engage, and interact with blog content seamlessly.

The page includes the following components:

Featured Image and Title Section: At the top, a visually appealing banner image is displayed along with the blog title (e.g., Python Tips). This enhances user engagement and provides context for the content.

Blog Content Area: The main section presents the full blog content in a structured and readable format. It may include paragraphs, code snippets, and bullet points to improve clarity and understanding.

Interaction Buttons: Users can interact with the post through various actions: Like/Unlike Button: Allows users to express appreciation for the post. Share Button: Enables sharing the blog across platforms. Edit Button: Available to the author for modifying the content. Delete Button: Allows the author to remove the post if needed.

Comments Section: This section enables user engagement and discussion: Comment Input Field: Users can write and submit their comments. Post Comment Button: Submits the comment to the system. Comment Display Area: Shows existing comments or a message if no comments are available.

Related Posts Section: Displays suggestions of similar or relevant blog posts based on category, tags, or user preferences. This supports content discovery and keeps users engaged within the platform.

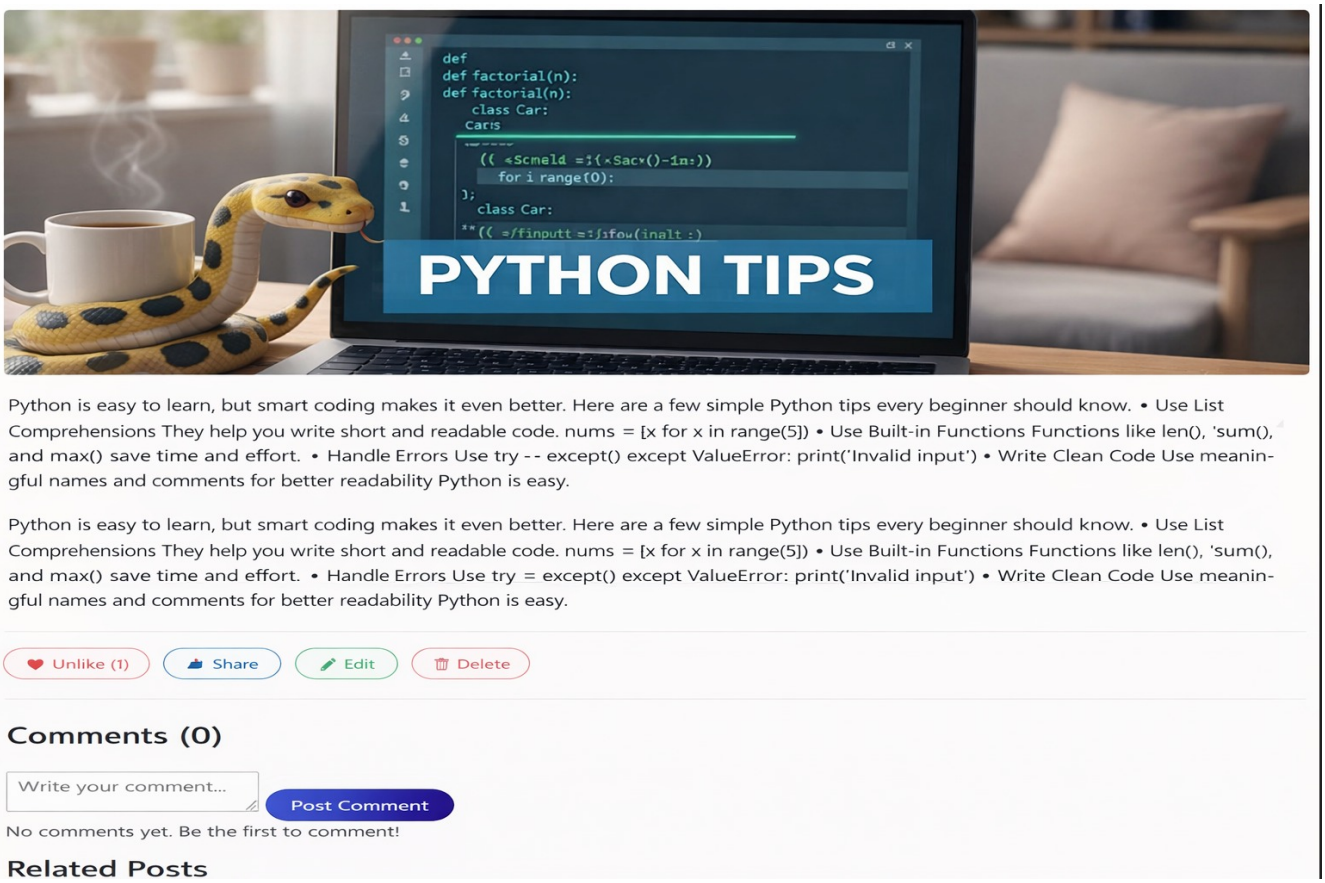

Figure 3.9: Blog Post Page

Chapter 4

Results / Analysis / Inferences

4.1 Results

- The blog website with KNN-based recommendation system was successfully developed and implemented. All core functionalities such as user registration, blog creation, editing, and viewing were working efficiently.
- The recommendation system provided relevant blog suggestions based on user interests. The system was tested and found to be reliable and user-friendly. It improved content discovery and overall user engagement. The project achieved its objectives successfully.
- The system provides a clean and intuitive user interface for easy navigation. Users can interact with the platform without any confusion. The responsive design ensures accessibility across devices. This enhances overall user satisfaction
- The website performed reliably during testing with minimal errors. It handled multiple operations efficiently without performance issues. The project met all its defined objectives successfully. It also demonstrates scalability for future enhancements.

4.2 Analysis

- The system was analyzed based on its core functionalities such as user registration, login, and blog management. All modules were tested individually and worked as expected. The integration between frontend and backend was smooth. The system successfully handled CRUD operations without errors.

- The system processes employee data such as job role, experience, salary, and satisfaction level to identify patterns related to attrition. During analysis, it was observed that factors like low job satisfaction, high workload, and fewer growth opportunities significantly influence an employee's decision to leave.

- The machine learning model demonstrates good predictive capability, providing reliable results when trained with properly preprocessed data. The use of data preprocessing techniques improves accuracy and ensures better model performance.

- The interface was analyzed for usability and accessibility. The layout was simple, clean, and easy to navigate for all users. The responsive design ensured proper functioning across different devices. Overall, users experienced smooth interaction with the system.

- The system was tested under different conditions to check its performance. It responded quickly to user actions and handled data efficiently. No major errors or delays were observed during testing. The system proved to be reliable and efficient for real-time use.

4.3 Inferences

- The blog website successfully combines web development with machine learning concepts.

- It demonstrates practical implementation of a recommendation system.

- The KNN algorithm provides relevant and personalized blog suggestions. This helps users find useful content easily.

- The system improves user engagement and overall experience. It makes content discovery faster and more efficient.

- All modules work smoothly and reliably together. The system is user-friendly and suitable for real-world use.

Chapter 5

Conclusion

The "Design and Development of a Blog Website with KNN-based Recommendation System" was successfully completed as per the planned objectives. The project effectively integrated frontend, backend, and database components to create a fully functional blogging platform. All the core modules, including user registration, login, blog management, and profile handling, were implemented and tested successfully.

The incorporation of the KNN-based recommendation system enhanced the functionality of the website by providing personalized blog suggestions to users. This feature improved content discovery and increased user engagement. The system demonstrated reliable performance, user-friendly interface, and efficient handling of data.

Overall, the project provided valuable practical exposure to real-world web development and machine learning applications. It helped in improving technical skills, problem-solving abilities, and understanding of system design. The developed system can be further enhanced with advanced features, making it suitable for real-time applications.

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